

ARUN ARUNACHALAM

Jersey City, NJ | 908-330-1050 | asnarun75@gmail.com | linkedin.com/in/arunarunachalam75

PROFESSIONAL SUMMARY

Data engineering executive with 17+ years architecting and modernizing enterprise data platforms across capital markets and financial services, with deep expertise in cloud-native data infrastructure, end-to-end data engineering, and platform transformation. Proven ability to define and enforce data standards, lead requirements gathering, and partner with business owners and technology leads to deliver scalable data processing and management solutions. Experienced in migrating legacy data systems to cloud platforms, building event-driven pipelines, and enabling AI-driven analytics across distributed, high-volume environments. Combines hands-on architectural depth in Spark, Databricks, Kafka, and AWS with executive stakeholder leadership to drive measurable business outcomes.

SIGNATURE IMPACT

- **\$8M** enterprise transformation → **40% performance improvement, 30% cost reduction**
- **AI-assisted engineering** → **35% faster delivery, 45% productivity gain**
- Scaled platforms from 10 to **50+ systems** with **99.97% uptime**
- Enabled **~\$2B capital efficiency** via risk platform transformation

CORE CAPABILITIES

Data Engineering (Spark, Databricks, ETL/ELT, Snowflake) | Cloud-Native Architecture (AWS, Kubernetes) | Data Modeling, Sourcing & Governance | Event-Driven Systems (Kafka) | Platform Modernization & Migration | AI & GenAI Platforms (RAG, Agentic AI) | Standards Definition & Enforcement | Executive & Business Stakeholder Partnership | Regulatory & Compliance Systems | Global Engineering Leadership

PROFESSIONAL EXPERIENCE

Bank of America — Director, Global Risk & Capital Markets Technology (2015 - Present)

- Architected and led end-to-end data engineering platforms supporting **\$1T+ daily processing**, encompassing sourcing, modeling, transformation, and delivery across distributed systems
- Defined and enforced data architecture standards across cloud-native, event-driven platforms (AWS, Kafka, Kubernetes), ensuring consistency and quality at enterprise scale
- Led requirements gathering and design sessions with business owners and technology leads to align data platform capabilities with trading, risk, and regulatory reporting needs
- Modernized legacy data infrastructure into cloud-native, event-driven architecture, improving data throughput by **40%** and reducing operational costs by **30%**
- Implemented AI-assisted development workflows and ML-based anomaly detection to improve data pipeline reliability and operational intelligence
- Ensured regulatory compliance (FRB, FINRA, SEC) across mission-critical data and risk systems
- Built and led global engineering teams (**30+ across US/UK/India**) driving **15% velocity improvement** in data platform delivery

Bank of America — VP, Structured Products & Mortgage Technology (2011 - 2015)

- Led consolidation and re-engineering of risk and analytics data systems, contributing to ~\$2B capital efficiency through improved data accuracy and processing
- Built real-time pricing and analytics platforms for structured products and mortgage trading, integrating complex data sourcing and modeling requirements
- Partnered with business stakeholders to translate trading desk requirements into scalable data engineering solutions

Bank of America — AVP, Whole Loan Trading Desk Support (prior to 2011)

- Architected and delivered end-to-end data and application pipelines supporting whole loan trading operations
- Introduced CI/CD practices reducing release cycle from monthly to weekly, improving data platform delivery cadence

Senior Programmer Analyst — Knight Trading (2004)

DB2 / WebSphere Administrator & Sr. Java Developer — Kyocera Mita (2003)

Sr. Java Web Developer — Mellon Investor Services (2001)

Sr. Java GUI Developer — Thomson Financials (2001)

Sr. Java GUI Developer — Icor Brokerage (2000)

AI & INNOVATION

- Building GenAI platforms (RAG pipelines, agentic AI systems) to augment data engineering workflows and enable intelligent decision systems
- Applied ML for anomaly detection and operational intelligence across large-scale data pipelines
- Driving AI-assisted SDLC workflows improving data platform delivery speed and quality

EDUCATION

Bachelor of Technology — Anna University

SKILLS

Architecture & Data Engineering: Cloud-native architecture (AWS, Kubernetes), event-driven systems (Kafka), distributed systems, enterprise data modeling, ETL/ELT pipeline design, data sourcing and governance, high-availability platforms (99.97% uptime) **Data & AI:** Spark, Databricks, Snowflake, Hadoop, Cassandra, MongoDB, GenAI (RAG pipelines, agentic AI), ML anomaly detection, AI-assisted SDLC **Engineering:** Java, Python, Node.js, SQL, Oracle, Sybase, REST/SOA, WebLogic, Kafka, Kubernetes, AWS **Leadership:** Executive stakeholder management, business partnership, standards definition and enforcement, global team leadership, P&L ownership, strategic planning, program management, Agile, ITIL, SOX, FRB/FINRA/SEC compliance **Tools:** ServiceNow, JIRA, Git, Visio, Microsoft Project